25900242 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCBTS200 - ESSENTIALS IN BIOLOGICAL TECHNIQUES

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Fill in the blanks

Answer all questions

Each question carries **1** marks

1. The first step in bioanalysis is _____ collection. [K] / [CO1]
2. Bioanalytical techniques are essential for _____ analysis of drugs and biomolecules. [K] / [CO1]
3. One advantage of the Ag/AgCl electrode is its _____. [K] / [CO2]
4. The Beer–Lambert’s law is also known as the law of _____. [K] / [CO2]
5. One limitation of a _____ is its limited wavelength range. [K] / [CO2]
6. The principle of electrophoresis is based on the migration of charged particles when an ____is applied. [U] / [CO2, CO3]
7. ____in the electrophoresis buffer acts as a chelating agent that can bind cofactors of DNA ase enzymes. [U] / [CO2]

8. The process of breaking cells to release cell organelle before centrifugation is called ___ [U] / [CO2]

9. …….....radiation of low energy can be measured by Liquid scintillation counter. [U] / [CO4]

10. …….....is used to diagnose bone diseases. [U] / [CO4]

[1x10 = 10]

Part B

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

11. Define colorimeter. [K] / [CO2]

12. Define ion-selective electrodes and give an example. [K] / [CO2]

13. Define pH and write its formula. [K] / [CO2]

14. State the advantages of a double cell over a single cell colorimete [K] / [CO2]

15. Give two practical uses of Beer–Lambert's law. [K] / [CO2]

16. Explain the significance of using Beta-mercaptoethanol in SDS -PAGE? [K] / [CO2, CO3]

17. Cite any 2 advantages of TLC over paper chromatography? [U] / [CO2]

18. Comment on any 2 application of ion exchange chromatography. [U] / [CO2, CO3]

19. Mention any 2 application of SEC in protein purification ? [K] / [CO2]

20. Cite the importance of centrifugal force in seperating cellular components . [U] / [CO2]

21. What is radioimmunoassay? [U] / [CO4]

22. Explain the principle of incorporation of radioisotopes into biological tissues and cells. [U] / [CO4]

[2x10 = 20]

Part C

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

23. Explain the applications of I-131 in medicine. [U] / [CO4]

24. Explain the major parts of a spectrophotometer and illustrate them with a diagram. [U] / [CO2]

25. Why is the glass electrode selective to hydrogen ions? [U] / [CO2]

26. Write a note on isoelectric focusing and its importance in protein characterization. [U] / [CO2, CO3]

27. Which analytical method you would prefer to seperate the individual components of a mixture? [U] / [CO2, CO3]

28. Enumerate the applications of radioisotopes in biology. [U] / [CO4]

29. Explain the application of Sr-85 in bone imaging. [U] / [CO4]

[4x5 = 20]